

Produce Once, Utilize Twice for Anomaly Detection

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Abstract—Visual anomaly detection aims at classifying and locating the regions that deviate from the normal appearance. Embedding-based methods and reconstruction-based methods are two main approaches for this task. The embedding-based methods typically predict the anomaly by measuring the distances between the deep representations of the test samples and a limited number of nominal samples, which enables these methods to be efficient but struggle in providing a fine-grained pixel-level anomaly location. The reconstruction-based methods rely on the pixel-level reconstruction errors to locate the anomaly, thereby the anomaly predictions are fine-grained. However, there are repetitive feature extractions and usually extra modules to guarantee the quality of the reconstructed images, resulting in unsatisfactory detection efficiency. In a nutshell, the prior methods are either not efficient or not precise enough for the industrial detection. To deal with this problem, we derive POUTA (Produce Once Utilize Twice for Anomaly detection), which improves both the accuracy and efficiency by reusing the discriminant information potential in the reconstructive network. We observe that the encoder and decoder representations of the reconstructive network are able to stand for the features of the original and reconstructed image respectively. And the discrepancies between the symmetric reconstructive representations provides roughly accurate anomaly information. To refine this information, a coarse-to-fine process is proposed in POUTA, which calibrates the semantics of each discriminative layer by the high-level representations and supervision loss. Equipped with the above modules, POUTA is endowed with the ability to provide a more precise anomaly location than the prior arts. Besides, the representation reuse also enables to exclude the feature extraction process in the discriminative network, which reduces the parameters and improves the efficiency. Extensive experiments show that, POUTA is superior or comparable to the prior methods with even less cost. Furthermore, POUTA also achieves better performance than the state-of-the-art few-shot anomaly detection methods without any special design, showing that POUTA has strong ability to learn representations inherent in the training data.

Index Terms—Anomaly detection, detection efficiency, representations reusing.

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I. INTRODUCTION

UNSUPERVISED anomaly detection has drawn much attention in recent years [26], [32], [54], since it requires no defect samples for training [25], [29], [31], which benefits the scenario where the collection of defect samples is costly, e.g. industrial detection [28], [30]. There are two main approaches for anomaly detection, reconstruction-based methods and embedding-based methods. The reconstruction-based methods train a reconstructive network to build a normal version of the input [27], as shown in Figure 1. Generally, the deviation lies solely in the abnormal regions before and after reconstruction. By analyzing the reconstruction errors, the anomaly can be located [2], [49], [54]. As illustrated in Figure 2, due to the analysis on pixel-level reconstruction errors, the reconstruction-based methods are endowed with the capability to locate the anomaly at the fine-grained pixel-level. However, there are strict requirements for the reconstructive network. Specifically, abnormal regions should be repaired to normal, while the normal regions are supposed to remain unchanged. Otherwise, the reconstruction errors will not conform to the anomaly, resulting in failure in detection, as shown in the carpet of Figure 2. Prior studies [6], [33] always design extra modules for the reconstructive network to meet the aforementioned requirements. Due to the intricate structure, the reconstruction-based methods fall flat when comparing the inference time with the embedding-based methods. The embedding-based methods typically exploit the deep representations to do a feature matching between the test sample and the nominal samples, as indicated in Figure 1. Since the nominal samples for feature matching are usually limited in number [16], the embedding-based methods achieve tremendous progress on the detection efficiency. However, the deep representations are typically of lower resolution than the input, which makes the embedding-based methods to struggle in providing a fine-grained pixel-level anomaly location, as shown in Figure 2. In a nutshell, both of the above methods have some deficiencies, and it is an ambitious goal for the prior arts to provide a fine-grained pixel-level anomaly location efficiently.

There are two approaches to overcome the aforementioned deficiencies. For the embedding-based methods, to obtain the fine-grained location, it is necessary to analyze the shallow representations to add details about the anomaly. RD [18] follows this way, which utilizes a distillation mechanism and introduces a student network to upsample the features from the teacher network to measure the shallow representations, as depicted in Figure 1. Specifically, RD believes the teacher network which is pre-trained on other dataset (such as

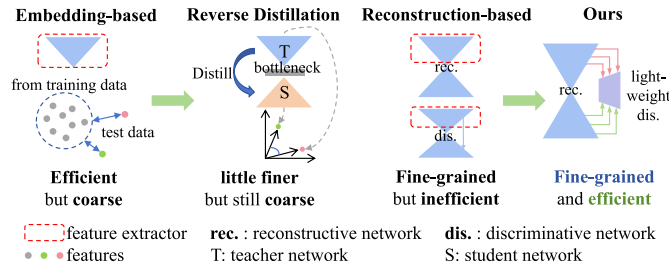


Fig. 1. The network architectures of the reconstruction-based method (DRaEM [5]), embedding-based method (PatchCore [16]), the improved embedding-based method (RD [18]) and the proposed POUTA which is improved based on the vanilla reconstruction-based method [5].

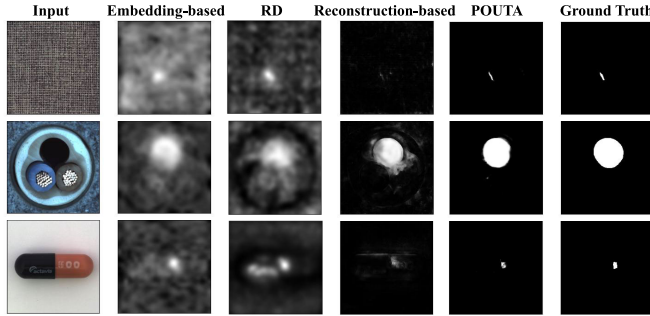


Fig. 2. The qualitative examples of the embedding-based method (PatchCore [16]), the improved embedding-based method (RD [18]), reconstruction-based method (DRaEM [5]), and the proposed POUTA.

ImageNet) has the ability to demonstrate the anomaly well, while the student network trained only on the current dataset does not have this ability. Therefore, the cosine distance between the features of the teacher network and student network reveals the anomaly location. However, modeling or feature matching the shallow representations is sophisticated. As shown in Figure 2, the predicted location of RD is finer than the embedding-based method, but as it identifies anomaly in the feature layer whose resolution is close to that of the input, the predicted anomaly location is not as fine-grained as that of the reconstruction-based methods. It is because that the shallow representations are not as homogeneous as the deep ones, it requires to record the shallow representations elaborately to obtain an accurate anomaly location, which is costly and may deprive the efficiency. Therefore, we turn to another solution, improving the accuracy and efficiency of the reconstruction-based methods.

Recent reconstruction-based methods follow a paradigm, which involves a reconstructive network and a discriminative network [5], [35], [48], as demonstrated in Figure 1. The discriminative network is employed to locate the anomaly by analyzing the discrepancies between the original and reconstructed images, demonstrating significantly better performance than simply measuring the absolute distance between these two images. Despite the impressive efforts, two critical issues still exist. Firstly, this paradigm yet relies on the reconstruction errors to identify anomaly. The requirements for the reconstructive network are strict as usual. While the quality of the reconstructed image is sometimes not so ideal, resulting in the unsatisfactory prediction, as shown in the carpet and

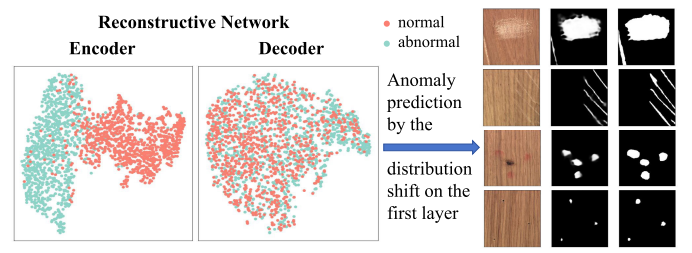


Fig. 3. In POUTA, the representation distributions of the last reconstructive encoder layer and the first reconstructive decoder layer are visualized by t-SNE [42]. The red points correspond to the representations of the abnormal regions, and the green ones refer to the normal representations. Each column on the right represents the input, the anomaly predicted only by the differences in representation of the aforementioned two layers, and the ground truth, respectively.

capsule of Figure 2. Even if the reconstructed image is of high quality, it is still a challenging problem to indicate anomaly completely when the color of abnormal region is close to the normal one. The second lies in the excessive computing consumption caused by the repetitive feature extraction in the reconstructive and discriminative network. Both of them extracts the features of the input in their encoders. It seems impossible to solve both of the above problems simultaneously within the conventional reconstruction-based paradigm, as achieving better performance necessitates enhancing the quality of reconstruction, which inevitably leads to increased computing consumption. This enlightens us to propose a new reconstruction-based paradigm.

We argue that, in the reconstructive network, the encoder and decoder representations are able to stand for the hierarchical representations of the input and reconstructed images. It is more suitable for the discriminative network to locate the anomaly by analyzing the discrepancies between the symmetric representations in the encoder and decoder directly. To verify our assumption, the encoder and decoder representations are visualized in Figure 3. Figure 3 indicates that, in the encoder, most of the abnormal representations are apart from the normal ones. While in the decoder, since the abnormal regions are repaired to normal, their representations are shifted to the distribution of the normal data. The above phenomena are consistent with our assumption. Since the normal regions and the abnormal regions in the input have different appearances, they naturally belong to different distributions. While the abnormal regions are repaired to normal in the reconstructed image, their representations will shift to the normal distribution in the decoder. Therefore, it is feasible to identify the anomaly by the representation discrepancies between encoder and decoder. As shown in Figure 3, by analyzing the disparities between the last reconstructive encoder layer and the first reconstructive decoder layer, the anomaly can be located roughly accurately.

Based on the above conclusion, we derive a neat and effective approach, POUTA (Produce Once Utilize Twice for Anomaly detection), as an effective remedy for the reconstruction-based methods on both accuracy and efficiency, fully reusing the discriminant information potential in the reconstructive network. POUTA is established with a view

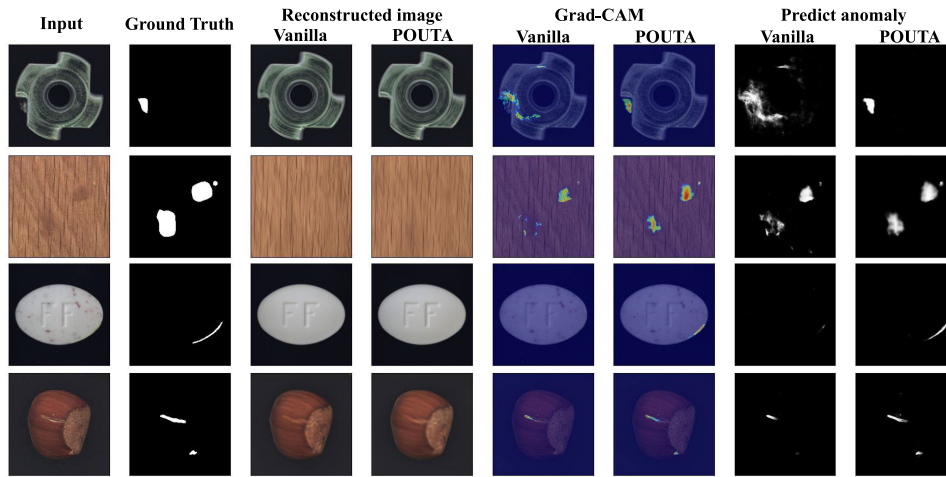


Fig. 4. The qualitative examples and visualizations (provided by Grad-CAM [4]) of the vanilla reconstruction-based (DR/EM [5]-like) and POUTA.

to the availability of the multi-scale reconstructive representations, aggregating hierarchical symmetric representations to identify anomaly through a lightweight coarse-to-fine discriminative network, as shown in Figure 1. Specifically, in the discriminative network, firstly, there is a MSS (Multi-scale Semantic Supervision) module to ensure clear differentiation for the abnormal regions between encoder and decoder representations. Secondly, a differential module named FCM (Feature Contrast Module) is designed to analyze the distribution shift of the same representation between the encoder and decoder in the feature space. The representations of the normal regions remain in the same distribution, while those of the abnormal ones shift, which can serve as a signal for detection. In addition, HSG (High-level Semantic Guidance) module calibrates the hierarchical representations output by FCM progressively, resulting in rich and precise spatial details. It can be observed that our method compares the shallow representations in the encoder with their symmetric decoder representations. While the anomaly information drawn from the discrepancies between these representations is calibrated by the deep semantics through the HSG module and also the MSS module, which enables the prediction to be more accurate and fine-grained simultaneously.

Due to reusing the reconstructive representations, POUTA suffices to identify anomaly more precisely than the vanilla reconstruction-based method with even lower cost. We take Grad-CAM [4] to visualize the semantic differences captured by the vanilla method and POUTA in Figure 4. There is no significant discrepancy between the reconstructed images from both methods, while POUTA obtains more complete semantics of anomaly regions through the representations and therefore the predictions of POUTA are more precise. This indicates that the reconstructive representations are more informative than reconstruction errors. It is reasonable since the information loss occurs when reconstructing an image from its representations. As a result of this loss, the pixel-level reconstruction errors might be incomplete, or even inaccurate, potentially leading to detection failures in certain cases, especially those of fine-grained anomalies. As shown in Figure 4, the colors of the abnormal regions in wood and pill are close to the

normal ones, the vanilla method fails to locate anomalies without surprise, but POUTA still identifies anomaly precisely. Besides, the discriminative network in POUTA directly reuses the reconstructive representations, which distinctively allows for the exclusion of feature extraction in the discriminative network, thereby improving efficiency. In summary, by directly reusing informative representations, POUTA relaxes the strict requirement for reconstructive network, and this new paradigm is able to achieve higher accuracy at lower cost.

Our main contributions can be listed as follows:

- Different from previous reconstruction-based methods, the discriminative network in POUTA directly reuses the representations in the reconstructive network, which filled the gap that the previous arts are typically inefficient.
- POUTA utilizes the discrepancies between the reconstructive representations to locate the anomaly, which provides more accurate and complete anomaly information than the previously used reconstruction error, endowing the capability to identify the anomaly more precisely.
- Experimental results show the superiority of the proposed unsupervised method POUTA. It outperforms the state-of-the-art methods on MVTec AD and VisA dataset and achieves a comparable performance with the best-performing supervised method on DAGM dataset.
- POUTA also achieves superior performance on few-shot anomaly detection without any special design, which indicates that POUTA has strong ability to learn representations inherent in the nominal data.

II. RELATED WORK

Anomaly detection is an unsupervised detection task, where only the normal data is involved amid training [58]. The anomaly detection model is required to distinguish between samples being drawn from the training data distribution and those outside its support [16], [61]. Given respective nominal samples and test samples, anomaly detection can be converted to calculating distances between the nominal deep representations and the test ones, or analyzing the differences before and after reconstruction, which corresponds to the embedding-based methods and reconstruction-based methods.

A. Embedding-Based Methods

The embedding-based methods typically extract the deep representations and then calculate the distance to k nearest nominal neighbors or the probability in the distribution. DeepSVDD [39] and MLBFD [59] propose to find the data centers and set corresponding radii. Data where the distance from its corresponding center exceeds the radius is considered as an anomaly. This approach is effective in fault diagnosis task. While DeepSVDD extracts the deep representations describing the entire image, which limits the model to obtaining only the existence of the anomaly but not to locating it. To mitigate this problem, PatchSVDD [3] extracts the deep representations for the image patch, which provides the rough anomaly location. Similarly, PaDim [40] generates patch embeddings from the pretrained network to estimate the Gaussian distribution moments (mean and covariance) for patch-level Mahalanobis distance. GLCF [28] points out that the performance of prior arts are limited since only the local knowledge is relied to identify anomaly. Therefore, GLCF proposes a two-branch approach to extract the local and global features, respectively. Despite improving the performance, the efficiency degrades. Other distribution learning methods can also be utilized, such as the normalizing flow (NF) [41]. DifferNet [36] takes the NF to measure the anomaly. To obtain the more informative representations, CSflow [37] builds a cross-scale NF, which enables the model to predict the anomaly more precisely while increase the model scale at the same time. Some works turn the anomaly detection task into a matter of distances to k nearest neighbors. SPADE records the nominal representations extracted from a pretrained network in a memory bank, and then calculates the average distances between the test data and k nearest nominal neighbors [15]. To notably reduce the inference cost, PatchCore [16] subsamples the memory bank. Moreover, Patchcore extracts locally aware patch features to achieve high performance. The traditional embedding-based methods suffer from the coarse anomaly prediction. RD [18] proposes to utilize a distillation mechanism into the embedding-based method. A student network is employed to upsample the representations of the teacher network, allowing to modelling the shallow representations whose resolution is close to the input. This approach slightly alleviates the difficulty of modelling the shallow representations, but also results in a predicted anomaly map that is not as fine-grained as that of the reconstruction-based methods. Unlike RD, we choose a different solution to consider both the efficiency and accuracy, improving the reconstruction-based methods by reusing the reconstructive representations. In this paper, our POUTA is compared with PatchCore and RD, known for their efficiency and finer location, respectively, among the above methods. The embedding-based methods generally enjoys higher efficiency than the reconstruction-based methods, but the anomaly locations are generally not as fine-grained as the reconstruction-based methods.

B. Reconstruction-Based Methods

The reconstruction-based anomaly methods reconstruct a normal version of the input, typically through the

autoencoder [10]. Since the anomaly is located by analyzing the pixel-level differences between the images before and after reconstruction, the anomaly prediction is always fine-grained. The early arts predicts the anomaly by the reconstruction error, typically in L1 distance [26], [27]. EaLDL [60] employs the L2 distance to calculate the reconstruction error. A threshold is set to distinguish normal and abnormal regions. Including the reconstruction error, SSR-AE [62] also calculate the L2 distance between the applied and predicted transformation on the original image, in order to enlarge the gap of anomaly scores between normal and abnormal images. SSAGAN [63] applies Peak Signal-to-Noise Ratio (PSNR) to calculate the prediction error between the actual pixels and the predicted ones provided by a self-attentive predictor. Besides, a self-supervised rotation angle prediction task is proposed to enlarge detection errors for abnormal pixels. TAC-Net [64] transforms the instances and incorporates contrastive learning during the training stage. The anomaly score is also measured by PSNR. These methods are effective in certain practical applications, such as process monitoring and video anomaly detection. Recently, to meet the requirements of more complex and changing industrial scenarios, there is a paradigm shift towards analyzing the reconstruction errors using a discriminative network [5], [21], [35], [48], [51]. Namely, the vanilla framework comprises a reconstructive network and a discriminative network in series. The input of the discriminative network contains only the input image and the reconstructed image. However, the procedure that compresses the representations to build a reconstructed image inevitably results in information loss, which may ultimately lead to inconsistencies between the reconstruction errors and the anomalies. Lots of effort has been invested in enhancing the quality of the reconstructed image to ensure that the reconstruction errors provide more precise anomaly information. And most turns to incorporate an additional module into the reconstructive network, such as a memory bank [6], [10], [24], [27], codebook [21], or attention mechanism [26], [33]. Turning the reconstruction sub-task to the image inpainting task, where the input is gridded and then reconstructed in patch [11], [22], [23], [46], also facilitates the construction of a high-quality reconstructed image. It also proposes to utilize the mathematical expectation map of the training dataset as the supplement of normal appearance [33], aiming to assist the reconstructive network to build a high-quality normal image. Based on the paradigm composed of a reconstructive network and a separate discriminative network, NIGSF [25] provides an insight into the normal appearance through another normal image, which is beneficial for identifying semantic anomaly. But the unavoidable inconsistencies between the input and the introduced one in details might lead to misjudgment. Despite being a smaller model, the performance drops on certain metrics. The above methods aim to build a high-quality reconstructed image, which follows the vanilla idea to locate the anomaly by analyzing the pixel-level reconstruction errors. It is costly but still can not deal with certain anomaly, such as the color-close-to-normal one. Aiming at improving both the accuracy and efficiency, it is necessary to propose a new paradigm. Our method opts to reuse the reconstructive representations,

which provides more precise anomaly information, endowing our method to achieve better performance with less costs than the vanilla method.

C. Few-Shot Anomaly Detection

Few-shot anomaly detection aims to indicate the anomaly with only a few nominal samples as the training support [20]. TDG [9] incorporates a hierarchical generative module to model the distribution of multi-scale patches. To enhance the representations, TDG uses image transformations and optimizes discriminators to distinguish between real and fake patches, as well as between different transformations applied to the patches. The anomaly predictions are obtained by aggregating the patch-based votes of the correct transformations. Similarly, RegAD [20] applies the transformations as well, but on the features, learning the category-agnostic feature registration flexibly. Therefore, RegAD can detect anomaly in new categories with just a few normal images, without requiring fine-tuning [20]. MSMC [43] proposes to employ a global memory bank and an individual memory bank to record the features of the entire image and a single object respectively. The prior few-shot anomaly detection methods implement focused design for the few-shot context, whereas our method POUTA is initially designed for the typical anomaly detection. Despite this, POUTA still exhibits considerable performance on the few-shot anomaly detection, indicating strong learning ability towards the representations inherent in the training data.

III. METHOD

As depicted in Figure 5, instead of analyzing the pixel-level reconstruction errors in the vanilla method, POUTA predicts the anomaly by analyzing the discrepancies in the symmetric reconstructive representations. POUTA is a neat and effective framework which comprises two networks that we will describe in sequence: self-supervised learning with reconstructive network (III-A) and coarse-to-fine discriminative network (III-B).

A. Self-Supervised Learning With Reconstructive Network

Different from conventional methods, the reconstructed image does not need to repair the anomaly flawlessly, and it is sufficient to offer a tendency towards repairing it in POUTA. Therefore, no additional elements are added to produce a flawless reconstructed image, and the reconstructive network is requested to rectify the synthetic anomaly using contextual information, which has been proven to be an effective strategy to enhance the representational learning ability [12], [17], [53].

The synthetic abnormal image I_{in} is generated by seamlessly blending an augmented image I_{aug} within a predetermined region M of an image I_{ori} from the training dataset.

$$I_{in} = I_{ori} \odot (\mathbf{1} - M) + I_{aug} \odot M \quad (1)$$

$$I_{aug} = \beta \times g(I_{ori}) + (1 - \beta) \times I_{un} \quad (2)$$

where I_{aug} is formed by merging a transformed version $g(\cdot)$ of the original image I_{ori} (e.g. translation, rotation or identity) with an image I_{un} obtained from an unrelated dataset [13].

Perlin generator [44] is used to determine the abnormal region. M is a binary map, which indicates the regions on a random Perlin noise map that is greater than a random threshold between 0 and 1. M is a zero matrix when the input is a normal image, all the numbers in the Perlin noise map are smaller than the threshold at this time. This situation rarely happens, in general, the inputs are mainly images with synthetic anomalies. β is a random number between 0 and 0.8. The input in Figure 5 exemplifies the synthetic image.

There are three parts in reconstructive network: the encoder, the mapping layer, and the decoder. Given the input image, the encoder outputs hierarchy representations F_{Ei} ($i = 1, 2, 3, 4$). Then, the mapping layer maps the representation F_{E4} to the normal latent space, and outputs as F_M . Subsequently, the decoder samples up F_M and obtains the multi-scale normal representations F_{Di} ($i = 4, 3, 2, 1$). Finally, an image with normal appearance I_r is reconstructed. I_r is supposed to be the same with the original one I_{ori} . Therefore, the reconstructive loss L_{rec} is calculated between I_r and I_{ori} by the mean square error (MSE) and structure similarity index measure [34] (SSIM). As mentioned in [5], MSE is a regular reconstruction loss but assumes the independence between neighboring pixels. Following recommendations in [5], we add SSIM loss into the reconstruction loss.

$$L_{rec} = L_{MSE}(I_r, I_{ori}) + L_{SSIM}(I_r, I_{ori}) \quad (3)$$

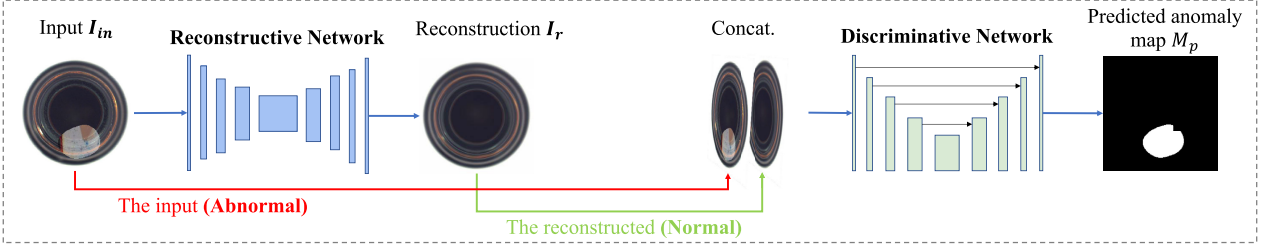
B. The Coarse-to-Fine Discriminative Network

The discrepancies between the symmetric reconstructive representations can accurately indicate the anomaly positions in some cases, as shown in Figure 2. While we find that the anomaly information contained in the reconstructive representations is coarse, which makes the indicated anomaly positions in other cases unclear or incomplete (we will present these cases in ablation study). Previous methods follow the vanilla paradigm to compress the representations into images. Nevertheless, the compression inevitably results in the loss of information, which leads to the pixel-level reconstruction errors not align with the anomalies. To address this, the previous methods focus on enhancing the quality of the reconstructed image by using sophisticated skills [6], [33], which typically requires extra space or computing resources.

Instead, the reconstructive representations are directly employed as discriminative representations in this paper. This approach eliminates the information loss associated with the compression from representations to images. To obtain a more precise result, we propose a coarse-to-fine strategy to analyze the discrepancies between the symmetric reconstructive representations. Three well-designed modules are proposed to detect and refine the anomaly location. Specifically, the Feature Contrast Module (FCM) initially acquires the coarse anomaly information within the reconstructive representations, which is the coarse stage. And then the High-level Semantic Guidance (HSG) and Multi-scale Semantic Supervision (MSS) refine the anomaly information, and identify the anomalies precisely.

1) *FCM*: FCM treats the encoder representations F_{Ei} ($i = 1, 2, 3, 4$) and the decoder representations F_{Di}

Utilize image-level reconstruction errors



Reuse reconstructive representations

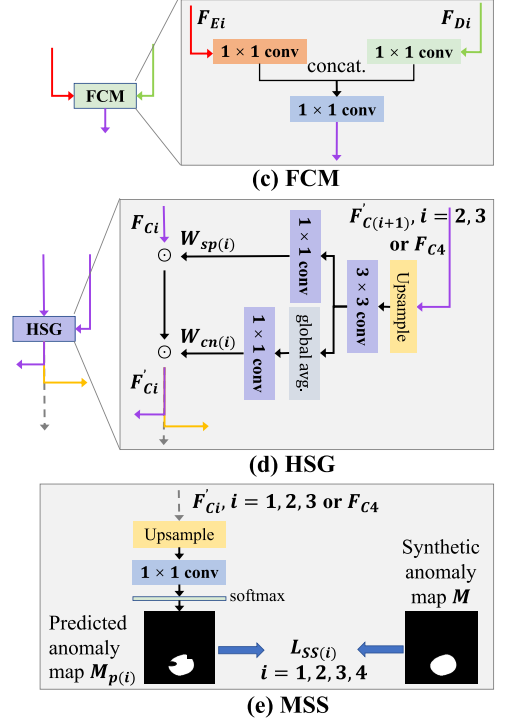
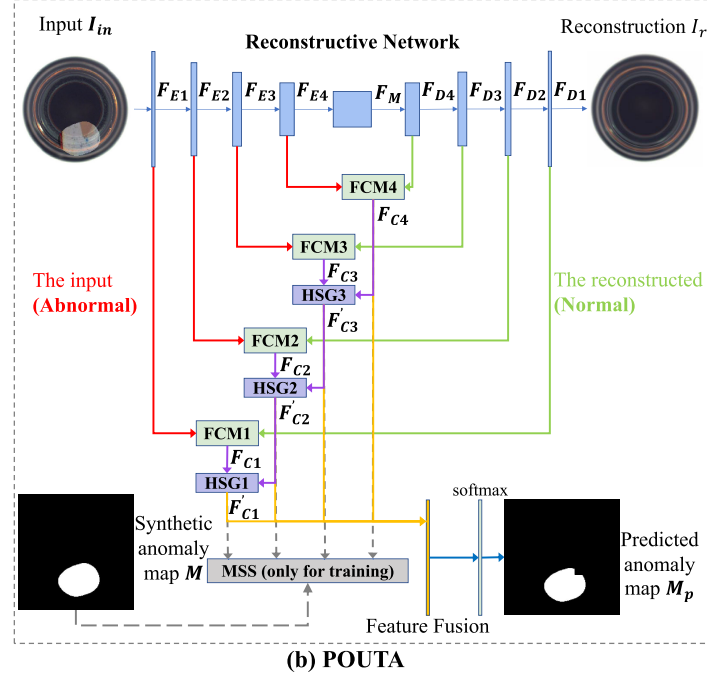


Fig. 5. The vanilla method and the proposed POUTA. Vanilla method is composed of an autoencoder-like reconstructive network and a U-Net [45]-like discriminative network, where the discriminative network obtains only the concatenation of the input and reconstructed image. While in POUTA, the discriminative network obtains the representations of the symmetric layers in the reconstructive network. The reconstructive representations contains more precise information about anomaly than the pixel-level reconstruction errors, which enables POUTA to identify anomaly more precisely. Each layer (except the first one) in the discriminative network involves a coarse (FCM module)-to-fine (HSG and MSS module) process.

($i = 1, 2, 3, 4$) as the features of the input and reconstructed image respectively. By analyzing the discrepancies between them, the preliminary anomaly information exists in the contrastive feature maps F_{Ci} ($i = 1, 2, 3, 4$). Specifically, as demonstrated in Figure 5, F_{Ei} and F_{Di} ($i = 1, 2, 3, 4$) go through their respective 1×1 convolutional layers. Next, a 1×1 convolutional layer is used to capture the discrepancies (distribution shift) between them by their channel-wise concatenation, and its output is denoted as F_{Ci} . F_{Ci} contains the roughly precise information about the anomaly location.

2) *HSG*: HSG aims to refine the anomaly information in F_{Ci} . And the refined version is denoted as F'_{Ci} ($i = 1, 2, 3$). Generally speaking, the shallow representations (high-resolution) contain the information about anomaly details, but they are also susceptible to interference from noise. And modelling the shallow representations is too sophisticated to be absolutely accurate, due to the various details in the shallow representations. Although the deep representations (low-resolution) generally cannot provide detailed anomaly information, they reveal accurate anomaly semantics and are less affected by noise. To refine the anomaly information

contained in F_{Ci} ($i = 1, 2, 3$), a natural idea is to filter the shallow F_{Ci} with the deep one $F_{C(i+1)}$.

As illustrated in Figure 5, the last refined representation $F'_{C(i+1)}$ ($i = 2, 3$ or F_{C4}) are used to filter the current representation F_{Ci} ($i = 1, 2, 3$). Specifically, the deep representation $F'_{C(i+1)}$ ($i = 2, 3$ or F_{C4}) is firstly upsampled to the current resolution, and then passes through a 3×3 convolution. Next, it generates weight maps at spatial dimension $W_{sp(i)}$ and channel dimension $W_{cn(i)}$ as filters. To generate the spatial weight map $W_{sp(i)}$, a 1×1 convolution is added to adjust the channel number to 1. As for the channel-wise weight map $W_{cn(i)}$, a global average pooling is used to adjust the spatial size to 1×1 , and then $W_{cn(i)}$ is generated by a 1×1 convolution. By multiplying $W_{sp(i)}$ and $W_{cn(i)}$ to F_{Ci} , a refined contrastive feature map F'_{Ci} is obtained:

$$W_{sp(i)} = \begin{cases} f_{sp(i)}(F'_{C(i+1)}), & i = 1, 2 \\ f_{sp(i)}(F_{C4}), & i = 3 \end{cases} \quad (4)$$

$$W_{cn(i)} = \begin{cases} f_{cn(i)}(F'_{C(i+1)}), & i = 1, 2 \\ f_{cn(i)}(F_{C4}), & i = 3 \end{cases} \quad (5)$$

$$F'_{Ci} = W_{sp(i)} \odot F_{Ci} \odot W_{cn(i)}, \quad i = 1, 2, 3 \quad (6)$$

where the $f_{sp(i)}$ represents the operations of upsample and convolution, and $f_{cn(i)}$ represents the operations of upsample, convolution and global average pooling. F'_{Ci} is involved to refine the next contrastive feature map $F_{C(i-1)}$ ($i = 2, 3$).

3) *MSS*: POUTA establishes on the observation that the representations of the encoder and decoder in the reconstructive network refer to the features of the input and reconstructed image respectively. Under this observation, we further propose a multi-scale semantic supervision (MSS) module to refine the anomaly information by ensuring a clear discrepancy between the symmetric reconstructive representations. MSS requires F'_{Ci} ($i = 1, 2, 3$) and F_{C4} to be able to predict the abnormal regions $M_{p(i)}$ through their corresponding predictive head. As shown in Figure 5, the predictive head consists of an upsample layer, a 1×1 convolution for dimensional reduction and a softmax layer. Note that MSS module is only used during training. To supervise the anomaly prediction of each layer, we add semantic supervision loss $L_{SS(i)}$ to each predictive head, which is represented as:

$$L_{SS(i)} = L_{focal}(M_{p(i)}, M) + L_1(M_{p(i)}, M), \quad i = 1, 2, 3, 4 \quad (7)$$

where $L_1(\cdot)$ denotes the ℓ_1 -norm-based loss function, $L_{focal}(\cdot)$ is the focal loss [7], M is a binary map for ground truth, which indicates the abnormal regions. It can be noticed that we use the L1-norm in Eq. 7 because it has been verified to be beneficial in segmentation more accurately, and retains less edge information than the L2-norm [52]. MSS forces the intermediate representations to pay more attention to anomaly information. The total loss of MSS L_{MSS} is the weighted sum of $L_{SS(i)}$ on each scale:

$$L_{MSS} = \sum_{i=1}^4 \lambda_i L_{SS(i)} \quad (8)$$

Finally, the channel-wise aggregation of F'_{C1} , F'_{C2} , F'_{C3} and F_{C4} is required to predict a pixel-level anomaly heatmap M_p . Inspired by the vanilla method [5], the image-level anomaly score S is the maximum of the 21×21 average pooling result of M_p . The loss of the predicted heatmap L_{pre} is measured by focal loss and L1 loss. L_{pre} presses the predicted anomaly map M_p to be as close to M as possible.

$$L_{pre} = L_{focal}(M_p, M) + L_1(M_p, M) \quad (9)$$

The total loss of POUTA L_{total} are represented as follows:

$$L_{total} = \alpha_1 L_{rec} + \alpha_2 L_{pre} + \alpha_3 L_{MSS} \quad (10)$$

L_{total} forces the reconstructed image I_r and predicted anomaly map M_p to be as close to I_{ori} and M as possible, respectively.

IV. EXPERIMENT

A. Experimental Setup

1) *Dataset*: POUTA is evaluated on MVTec AD [14], VisA [19], MPDD [68] and DAGM [47] dataset. MVTec

AD is a benchmark dataset for visual anomaly detection, which contains 5,354 images in 15 image categories. VisA contains 10,821 images with 9,621 normal and 1,200 abnormal samples. As for MPDD, it incorporates 1,346 samples in 6 classes. Since abnormal images only exist in the testing set for MVTec AD, MPDD and VisA, they are mainly used to train unsupervised anomaly detection methods. Different with the above datasets, DAGM has defect images in training set and is mainly used to train supervised anomaly detection methods. It contains 16100 images in 10 categories. Note that POUTA doesn't use any real defect images to train the model on all of the datasets, and it is only trained on the normal images in training data with the synthetic anomalies.

2) *Metric*: In order to have a fair comparison with other methods, both the image-level classification performance (CLS) and pixel-level location performance (LOC) are exploited in this paper. The metric to measure CLS and LOC is the area under the receiver operating characteristic curve (AUROC). We notice that many methods report high performance on MVTec AD, even close to 1, on AUROC metric. Therefore, we add another metric, average precision (AP) to measure LOC. On MVTec AD dataset, both CLS and LOC are evaluated. Since the annotations of DAGM is coarse, only the CLS is tested. Similar with other state-of-the-art methods, CLS is measured on VisA and MPDD.

3) *Implementation Details*: λ_1 , λ_2 , λ_3 and λ_4 in Eq. 8 are empirically set to 0.4, 0.3, 0.2 and 0.1 respectively. To confirm the reweight parameters α_1 , α_2 , and α_3 in Eq. 10, we conduct 25 groups of experiments as shown in Figure 6. We first adopt AP metric to evaluate the performance, due to AP is more sensitive to performance changes than AUROC. Since the performances of the current methods vary acutely on carpet, cable, pill, toothbrush, and transistor classes, as demonstrated on Table II, we thereby test the performance of the reweighted POUTA on these five objects. The distribution of the average performance on them is close to a normal distribution, the mean locates around $\alpha_1 = 1.0$, $\alpha_2 = 1.0$, $\alpha_3 = 1.0$, which is finally adopted in POUTA. POUTA is trained for 600 epochs with a batch size of 8. The learning rate of the Adam optimizer is set to 0.0002 and multiplied by 0.2 at epoch 480 and epoch 540. Images are resized to 224×224 .

4) *Training Algorithm*: Every experiment is trained from scratch, and initialized randomly, including the few-shot experiment. All the networks in POUTA is trained simultaneously as shown in Algorithm 1. L_{total} will force the reconstructive network to keep the features of the input in encoder, repair the features of the abnormal regions in mapping layer, and upsample the normal features in decoder. Regarding the discriminative network, L_{total} endows it with the ability to identify the discrepancies between the symmetric reconstructive representations. Figure 7 exhibits an example of L_{total} loss curve, which demonstrates that despite slight fluctuations, the overall tendency is a gradual decline that ultimately converges.

B. Anomaly Detection

The proposed POUTA is compared with previous methods on MVTec AD, VisA and DAGM. The comparison results are shown in Table I, II, III and V.

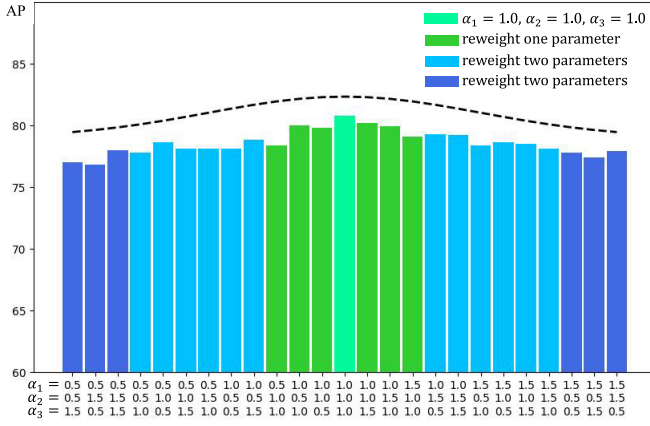


Fig. 6. The average performance of carpet, cable, pill, toothbrush, and transistor on AP metric. It can be seen that the best performance achieves when $\alpha_1 = 1.0, \alpha_2 = 1.0, \alpha_3 = 1.0$.

Algorithm 1 Training Algorithm

Data: original training image I_{ori}
Result: reconstructed image I_r , anomaly prediction in MSS
 $M_{p(i)}, i = 1, 2, 3, 4$, final predicted anomaly map M_p
random initialization
for $epoch \leftarrow 1$ **to** 600 **do**
 Synthesize abnormal image I_{in} from I_{ori} as in Equ. 1;
 Reconstructive network:
 compute reconstructed image I_r ;
 Calculate reconstruction loss L_{rec} as in Equ. 3;
 Discriminative network:
 Each FCM: compute $F_{Ci}, i = 1, 2, 3, 4$;
 Each HSG: compute $F'_{Ci}, i = 1, 2, 3$;
 MSS: compute $M_{p(i)}, i = 1, 2, 3, 4$;
 Calculate MSS loss L_{MSS} as in Equ. 8;
 Compute the final anomaly prediction map M_p
 Calculate prediction loss L_{pre} as in Equ. 9;
 Calculate total loss L_{total} as in Equ. 10;
 Back propagate L_{total} to optimize POUTA.
end

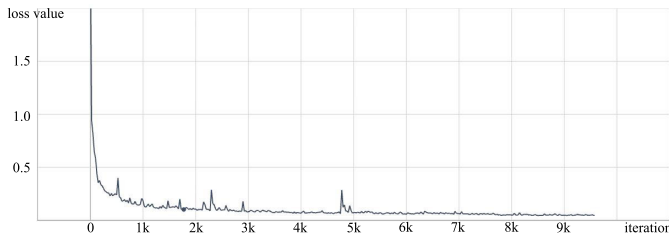


Fig. 7. The loss curve of L_{total} during training POUTA on connector.

1) *MVTec AD*: We report the quantitative classification (CLS) and location (LOC) results of the prior methods and POUTA on MVTec AD dataset in Table I and II, respectively. PatchCore in this section is conducted using its official code with 1% training data as coreset remaining. As depicted by Table I, POUTA shows highest classification performance on most cases, and outperforms the improved embedding-based method RD [18] in all classes except for metal nut. Especially on capsule, POUTA outperforms the runner-up by 1.4%. We also notice that the difference between POUTA and the best result is 2% on cable, which may due to changes in cable's

orientation, as well as the background. They prevent POUTA from modeling the normal appearance of cable, causing to judge these changes as anomalies. For the average performance of CLS, POUTA attains a comparable performance with SOTA results.

And for the LOC results in Table II, POUTA also demonstrates satisfactory performance in the majority cases, with little performance difference in the remaining cases. In addition, POUTA shows better average performance than the improved embedding-based method RD [18]. For the AUROC metric, POUTA surpasses the SOTA by 2.6% on transistor, which is because of the more complete anomaly location predicted by POUTA, especially when the anomaly is large. The performance improvement is more clear on AP metric. On wood, cable, and transistor, POUTA outperforms the SOTA by 6.9%, 9.1%, and 7.0%. On the average LOC results, POUTA is 0.3% and 2.9% higher than SOTA measured by pixel-level AUROC and AP metric respectively.

A qualitative comparison with the vanilla method (DRAEM [5]) and embedding-based method (PatchCore [16]) is presented in Figure 8, indicating that POUTA is able to precisely predict the anomaly with more details. The improved embedding-based method, RD [18] provides finer anomaly location than the embedding-based method PatchCore [16], while it is still cruder than the one of POUTA. It is worth mentioning that, Figure 8 shows POUTA is able to identify the minor anomalies, since the information about the minor anomalies can be delivered to POUTA by the discrepancies between the low-level reconstructive representations.

2) *VisA*: As shown in Table III, POUTA outperforms the prior methods on most cases. Especially on capsules and macaroni2, POUTA shows 21.0% and 17.1% improvement on the image-level AUROC metric. Both of these are the categories where the position of the object changes drastically per image. This indicates that POUTA is more flexible than others to adapt to these position changes. And as for the average classification performance, POUTA is 0.4% higher than the SOTA. Sample segmentations in Figure 9 offer a qualitative anomaly localization of POUTA, showing that POUTA can locate the anomaly accurately.

3) *MPDD*: Table IV depicts the quantitative results on MPDD. POUTA also outperforms others, including the improved embedding method RD [18], on most classes and average performance. This is particularly evident in the bracket brown and bracket white, where the color of the anomaly is often close to that of the normal one. While POUTA learns the anomalies semantically by the discrepancies between the reconstructive representations, which enables it to identify the color-close-to-normal anomalies. The only area for improvement lies in bracket black for 1.6% less than the SOTA on AUROC metric, as POUTA considers some changes in detail, such as the lighting changes as anomalies, as shown in column 2 of Figure 10. Other qualitative results of POUTA in Figure 10 indicates that POUTA can identify most of the anomalies accurately. The predicted maps in columns 4, 6, 7, 8, and 9 in Figure 10 are almost identical to the ground truth. In certain instances, such as columns 3 and 5, the ground truth labels are crude. While as a reconstruction-based

TABLE I
IMAGE-LEVEL CLASSIFICATION (CLS) (IN %) OF THE VANILLA METHOD DRÆM, SEVERAL STATE-OF-THE-ART RECONSTRUCTION-BASED METHODS, EMBEDDING-BASED METHOD AND OUR METHOD POUTA ON MVTEC AD DATASET. THE BEST RESULT FOR EACH CLASS IS HIGHLIGHTED IN BOLD

class	DRÆM[5]	RSTD[48]	ReContrast[55]	DBPI[33]	RD[18]	PatchCore[16]	POUTA
carpet	97.0	98.7	99.8	99.7	98.9	98.5	99.1
grid	99.9	100.0	100.0	99.8	100.0	98.2	100.0
leather	100.0	100.0	100.0	100.0	100.0	100.0	100.0
tile	99.6	99.9	99.8	100.0	99.3	99.1	100.0
wood	99.1	99.3	99.0	99.7	99.2	99.2	100.0
bottle	99.2	100.0	100.0	99.9	100.0	100.0	100.0
cable	91.8	99.6	99.8	95.7	95.0	99.3	97.8
capsule	98.5	93.0	97.7	97.1	96.3	96.7	99.9
hazelnut	100.0	99.8	100.0	100.0	99.9	100.0	100.0
metal nut	98.7	100.0	100.0	98.9	100.0	99.7	99.7
pill	98.9	98.1	98.6	96.6	96.6	98.0	98.0
screw	93.9	96.8	98.0	98.4	97.0	97.2	98.5
toothbrush	100.0	97.9	100.0	100.0	99.5	99.4	100.0
transistor	93.1	98.3	99.7	99.1	96.7	100.0	100.0
zipper	100.0	99.3	99.5	98.8	98.5	99.5	100.0
avg	98.0	98.7	99.5	98.9	98.5	99.0	99.5

TABLE II
PIXEL-LEVEL LOCATION (LOC) (IN %) OF THE VANILLA METHOD DRÆM, SEVERAL STATE-OF-THE-ART RECONSTRUCTION-BASED METHODS, EMBEDDING-BASED METHOD AND OUR METHOD POUTA ON MVTEC AD DATASET. THE BEST RESULT FOR EACH CLASS IS HIGHLIGHTED IN BOLD

class	DRÆM[5]		RSTD[48]		DeSTSeg[51]		DBPI[33]		RD[18]		PatchCore[16]		POUTA	
	AUROC	AP	AUROC	AP	AUROC	AP	AUROC	AP	AUROC	AP	AUROC	AP	AUROC	AP
carpet	95.5	53.5	99.2	-	96.1	72.8	98.7	80.6	98.9	-	99.0	63.9	98.0	77.2
grid	99.7	65.7	99.6	-	99.1	61.5	99.6	69.1	99.3	-	98.5	31.5	99.2	70.8
leather	98.6	75.3	99.6	-	99.7	75.6	99.4	76.0	99.4	-	99.3	46.2	98.4	71.0
tile	99.2	92.3	98.8	-	98.0	90.0	99.5	96.9	95.6	-	95.8	55.3	99.7	97.9
wood	96.4	77.7	98.1	-	97.7	81.9	96.9	78.5	95.3	-	95.1	50.1	98.2	85.4
bottle	99.1	86.5	99.3	-	99.2	90.3	98.9	88.0	98.7	-	98.5	77.5	99.3	91.6
cable	94.7	52.4	98.3	-	97.3	60.4	96.7	66.8	97.4	-	98.1	66.5	98.4	75.9
capsule	94.3	49.4	98.5	-	99.1	56.3	98.6	48.2	98.7	-	99.0	44.5	99.1	54.7
hazelnut	99.7	92.9	99.5	-	99.6	88.4	99.5	89.4	98.9	-	98.6	52.3	99.6	88.0
metal nut	99.5	96.3	98.9	-	98.6	93.5	98.0	92.2	97.3	-	98.3	87.0	99.3	97.1
pill	97.6	48.5	98.7	-	98.7	83.1	99.4	88.2	98.2	-	97.6	78.6	99.3	88.8
screw	97.6	58.2	99.3	-	98.5	58.7	99.3	60.7	99.6	-	99.4	37.2	97.7	58.1
toothbrush	98.1	44.7	99.3	-	99.3	75.2	99.3	78.1	99.1	-	98.5	37.4	99.5	82.8
transistor	90.9	50.7	90.7	-	89.1	64.8	95.3	72.5	92.5	-	95.1	65.0	97.9	79.5
zipper	98.8	81.5	99.2	-	99.1	85.2	98.6	74.1	98.2	-	98.8	62.3	98.9	83.5
avg	97.3	68.4	98.5	-	97.9	75.8	98.9	77.3	97.8	-	98.0	57.0	98.8	80.2

TABLE III
IMAGE-LEVEL CLASSIFICATION (CLS) (IN %) OF SEVERAL STATE-OF-THE-ART RECONSTRUCTION-BASED METHODS, EMBEDDING-BASED METHODS, AND OUR POUTA ON VISA DATASET (AUROC). THE BEST RESULT IS HIGHLIGHTED IN BOLD

	PCB1	PCB2	PCB3	PCB4	Capsules	Candle	Macaroni1	Macaroni2	Cashew	Chewing gum	Fryum	Pipe fryum	avg.
EdgRec[56]	-	-	-	-	-	-	-	-	-	-	-	-	94.2
D3AD[71]	-	-	-	-	-	-	-	-	-	-	-	-	96.0
ReContrast[55]	-	-	-	-	-	-	-	-	-	-	-	-	97.5
PatchCore[16]	98.5	97.1	98.4	99.8	76.2	98.9	97.2	78.6	97.9	98.5	94.5	99.7	94.6
RD [18]	-	-	-	-	-	-	-	-	-	-	-	-	96.0
POUTA	97.0	99.6	98.4	99.1	97.2	96.9	95.7	95.7	96.4	99.7	98.8	99.7	97.9

TABLE IV
IMAGE-LEVEL CLASSIFICATION (CLS) (IN %) OF SEVERAL STATE-OF-THE-ART METHODS AND OUR POUTA ON MPDD DATASET (AUROC). THE BEST RESULT IS HIGHLIGHTED IN BOLD

	bracket black	bracket brown	bracket white	connector	metal plate	tubes	avg.
RealNet[70]	95.0	96.8	88.8	100.0	100.0	97.5	96.3
PatchCore[16]	81.9	78.4	76.0	96.7	100.0	59.7	82.1
CFlow-AD[69]	81.9	78.4	76.0	96.7	100.0	59.7	82.1
PaDiM[40]	75.6	85.4	82.2	91.7	56.3	57.5	74.8
skip-GANomaly[49]	61.3	62.1	73.3	72.6	73.2	46.4	64.8
RD[18]	84.7	92.4	84.5	98.3	100.0	89.0	91.5
POUTA	93.4	97.8	95.3	100.0	100.0	98.5	97.5

method, POUTA provides a more fine-grained location in columns 3 and 5. Due to the varying shooting angles of the training images and the limited number of training images,

it is challenging to model the normal appearance in POUTA, leading to some misjudgments in the anomaly predictions for columns 3 and 5.

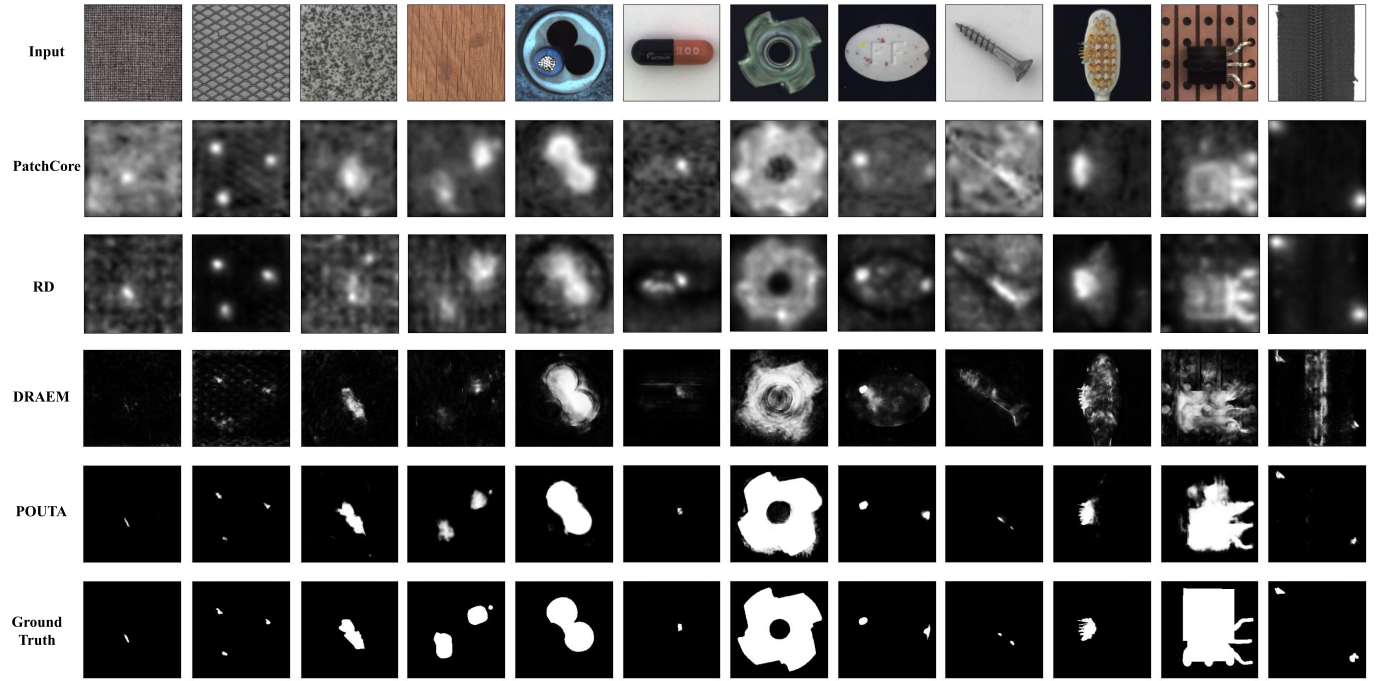


Fig. 8. The qualitative results of PatchCore [16], RD [18], DRAEM [5], and POUTA on MVTec AD [14].

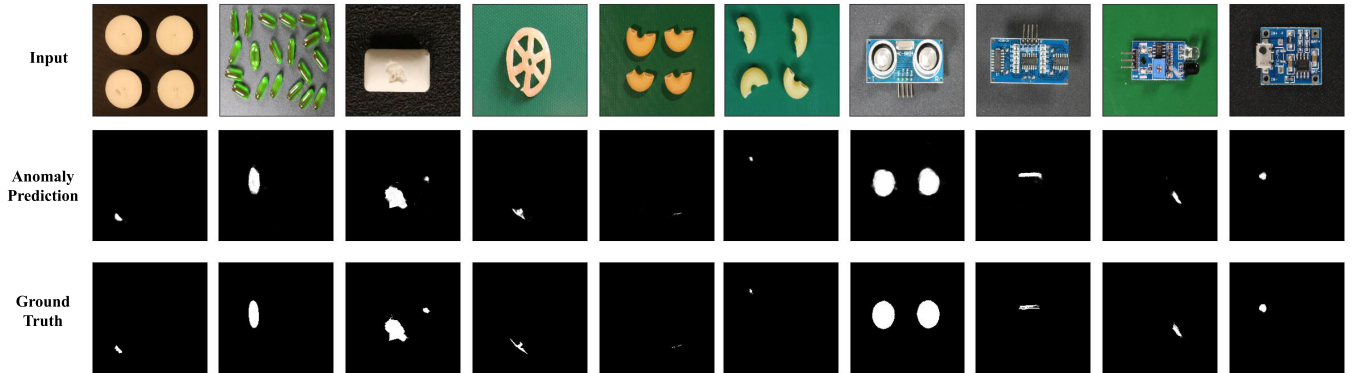


Fig. 9. The qualitative results of POUTA on VisA dataset [19].

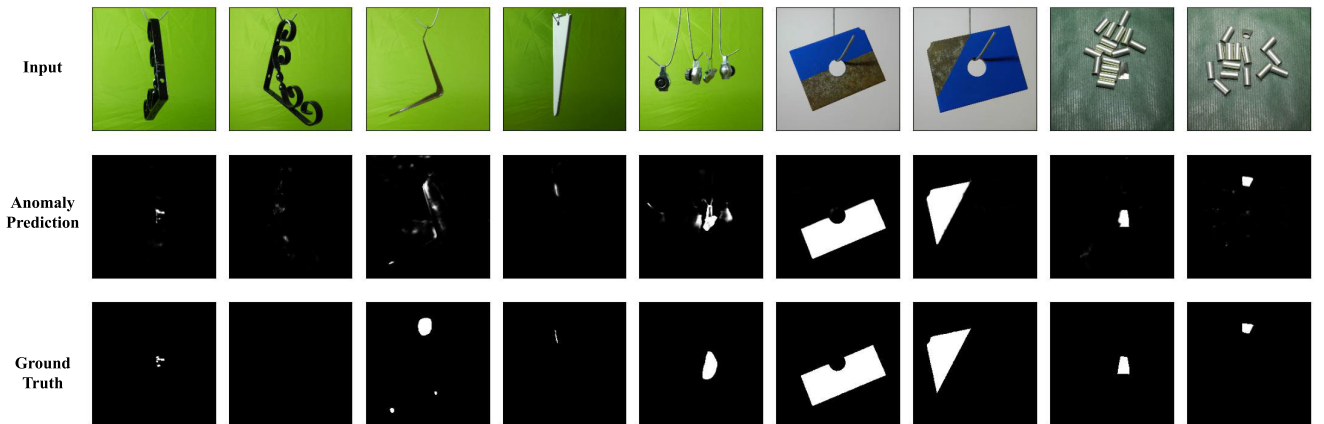


Fig. 10. The qualitative results of POUTA on MPDD dataset [68].

4) *DAGM*: In addition, the quantitative evaluations on DAGM dataset are presented in Table V. POUTA outperforms all the unsupervised methods, including the improved

embedding-based method, RD [18] on all the categories except for Class4. It is worth mentioning that on Class8, POUTA is 38.6% higher than PatchCore, which measures anomaly

TABLE V

IMAGE-LEVEL CLASSIFICATION (CLS) (IN %) OF SEVERAL UNSUPERVISED METHODS, SUPERVISED METHODS, AND OUR POUTA ON DAGM DATASET (AUROC). THE BEST RESULT IS HIGHLIGHTED IN BOLD

		Class1	Class2	Class3	Class4	Class5	Class6	Class7	Class8	Class9	Class10	avg.
unsup.	DBPI[33]	-	-	-	-	-	-	-	-	-	-	99.0
	DR/EM[5]	-	-	-	-	-	-	-	-	-	-	96.0
	PatchCore[16]	87.3	100.0	93.5	100.0	91.4	97.9	99.8	61.4	95.4	100.0	92.7
	RD[18]	92.4	99.6	82.3	100.0	79.3	98.8	99.7	58.8	94.2	99.6	90.5
	POUTA	100.0	100.0	100.0	99.9	99.9	100.0	100.0	100.0	97.4	100.0	99.7
sup.	CCNN[38]	99.0	100.0	98.0	99.0	100.0	100.0	100.0	100.0	100.0	100.0	99.6
	MS[57]	-	-	-	-	-	-	-	-	-	-	100.0

TABLE VI

IMAGE-LEVEL FEW-SHOT CLASSIFICATION (CLS) (AUROC IN %) OF THE PRIOR FEW-SHOT ANOMALY DETECTION METHODS (INCLUDING TDG [9], DIFFERNET [36], REGAD [20], AND POUTA ON MVTEC AD DATASET. THE BEST RESULT IS HIGHLIGHTED IN BOLD

	2-shot	4-shot	8-shot
TDG[9]	71.2	72.7	75.2
DifferNet[36]	80.5	80.8	82.9
RegAD[20]	81.5	84.9	87.4
POUTA	86.0	88.6	92.2

by calculating the distances between the test sample and the coreset's training samples. However, the texture of Class8 varies, which makes it to be impossible to include all possible textures in the training data, resulting in a non-ideal detection performance. While POUTA learns the distribution of the texture in the reconstructive network, which results in minimal texture changes before and after reconstruction, leading to more accurate detection results. The average performance is also higher than the prior unsupervised methods. Besides, POUTA achieves a comparable performance with the best supervised methods. While POUTA requires no real anomaly for training, which enables POUTA to be more flexible to the situation where the real anomaly is hard to collect.

C. Few-Shot Anomaly Detection

POUTA represents satisfactory performance on MVTEC AD with the entire training data. To assess the performance with a limited number of training images, we conduct a few-shot experiment. Few-shot anomaly detection becomes a hot topic recently since it is suitable for practical applications where the normal samples are hard to collect as well. Experiments are also done to analyze the effectiveness of POUTA on this task.

We train POUTA only with the support set (k training images, $k = 2, 4, 8$) for each category. All the settings keep the same with the ones in regular anomaly detection, we just reduce the number of the training images. Table VI demonstrates the quantitative results of TDG [9], DifferNet [36], RegAD [20] and POUTA on MVTEC AD. TDG and TEGAD are designed for the few-shot anomaly detection methods, and DifferNet is an embedding-based anomaly detection methods with the normalizing flow. Moreover, the qualitative results are represented in Figure 11.

Table VI indicates that POUTA achieves better performance than the prior few-shot anomaly detection methods. In 2-shot, 4-shot, and 8-shot, POUTA surpasses the run-up by 4.5%, 3.7%, and 4.8%, respectively. Whereas there is no special

design for the few-shot detection in POUTA. This is a surprising result, showing that POUTA has a strong ability to learn representations inherent in the training data.

Figure 11 demonstrates the qualitative results of POUTA for both the few-shot and regular anomaly detection. As shown in Figure 11, the obvious anomaly can be identified by POUTA even in the 2-shot setting. When rising the k to 8, the prediction for the obvious anomaly can rival the one provided by the model trained on the entire data, indicating that POUTA is able to learn a near accurate distribution of the normal data. As for the subtle anomaly, it is challenging for the few-shot model to precisely locate the anomaly. The location accuracy increases with the number of training images rising. The subtle anomaly can be partially predicted in some cases with the 8-shot setting, but it is still not as comprehensive as the predicted anomaly map output by the model trained on the entire dataset. In that the subtle anomaly tends to be close to the edge of the normal data distribution, definitely requiring more training data to construct an accurate distribution. In terms of the changable object, the prediction accuracy is obviously related to the number of the training images k . The qualitative result under $k = 8$ is close to the one under entire data training, and much more accurate than the one under $k = 2$ and $k = 4$. This is due to the fact that the more training images there are, the more the model acquires information about the object in various rotations, leading to a better understanding of its normal appearance.

The performance of POUTA under a few-shot setting can be summarized as follows: The performance is determined by whether the k training images can represent comprehensive information about the normal appearance to enable the model to clearly define the boundary between normal and abnormal. As for the obvious anomaly, it significantly differs from the normal appearance, making it easy to identify. Regarding the subtle anomaly, it is always on the edge of the normal data distribution and thereby close to the decision boundary. When k is small, it is challenging to fit the learned decision boundary closely to the real one, making it possible to take certain subtle anomaly as normal. When it comes to the changeable object, a few training images may not be sufficient to help the model fully understand the normal appearance. Hence, the anomaly prediction is not satisfactory when k is small, but the results markedly improve even with a few training images added (from $k = 2$ to $k = 8$).

The reconstructive representation distributions of POUTA in few-shot and regular anomaly detection are visualized in Figure 12. It can be seen that, similar with the distributions in

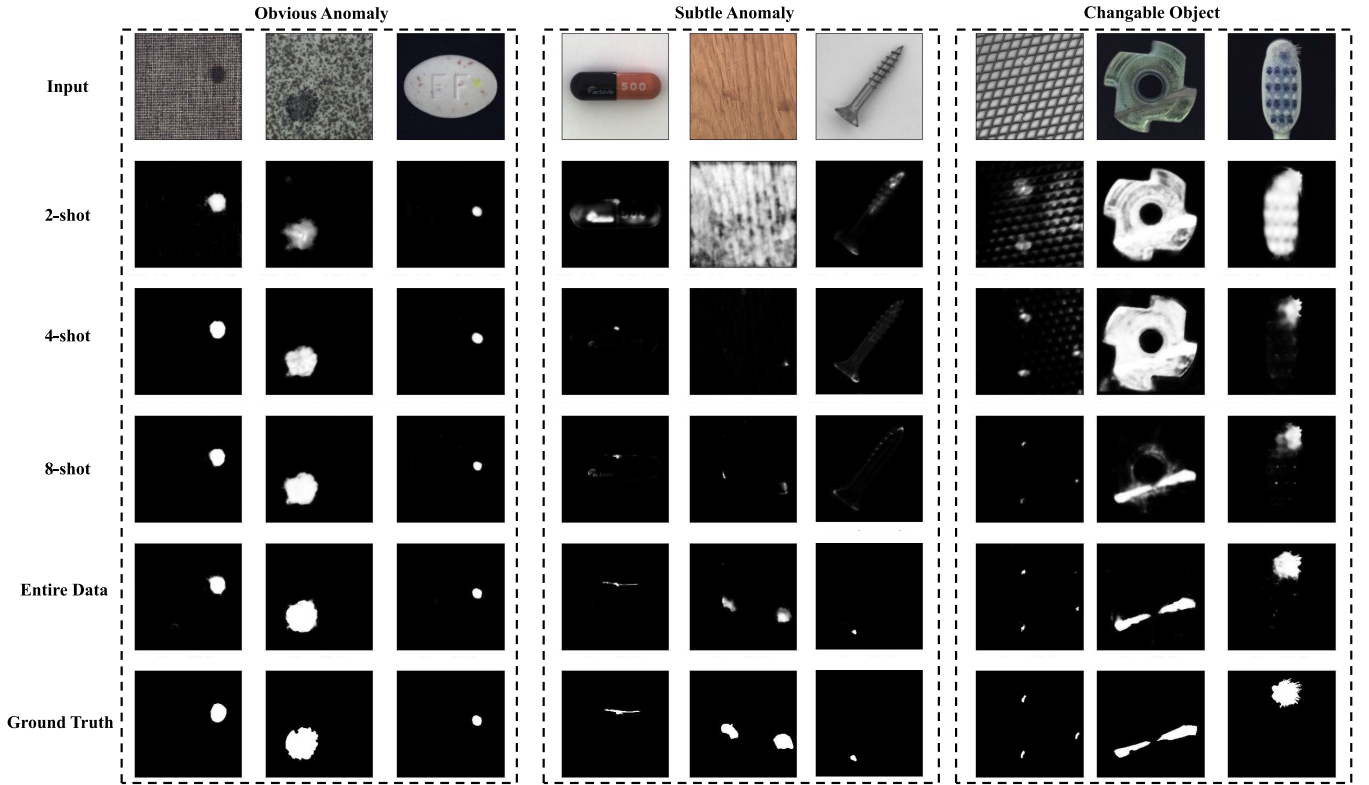


Fig. 11. The qualitative few-shot and regular anomaly detection results of POUTA on MVTec AD [14].

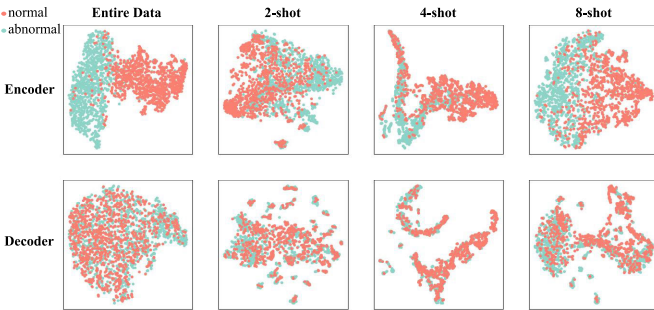


Fig. 12. The visualization of the encoder and decoder representations of POUTA through t-SNE [42]. The red points correspond to the representations of the abnormal regions, and the green ones refer to the normal representations.

regular anomaly detection, the distributions in the few-shot setting also follow the rule: In the encoder, the normal and abnormal representations are separated from each other (though not obvious), while in the decoder, they are distributed together as the abnormal representations have been repaired to normal. Besides, the visualization shows that with a few more training images (from 2-shot to 8-shot), the distribution established by POUTA is greatly improved, approaching to the distribution of the entire data training. For example, the normal and abnormal representations are separated more and more clearly from each other in the encoder. This confirms the strong ability to learn the inherent distribution of normal data.

D. Ablation Study

To investigate the effect of each component in POUTA, five groups of ablation experiments are conducted on MVTec AD. All the experiments share the same implementation details.

TABLE VII

ABLATION STUDY OF POUTA ON MVTec AD. THE BEST RESULT IS HIGHLIGHTED IN BOLD

Method	Module			average	
	FCM(Reuse)	HSG	MSS	CLS	LOC
Vanilla				98.4	98.1/75.4
POUTA-base	✓			98.8	98.1/76.7
POUTA-base+HSG	✓	✓		98.9	98.4/76.8
POUTA-base+MSS	✓		✓	98.8	98.3/77.2
POUTA	✓	✓	✓	99.5	98.8/80.2

1) *Quantitative Results:* The quantitative results are shown in Table VII. *Vanilla* refers to the vanilla method as shown in Figure 5 (a), which utilizes only the pixel-level reconstruction errors to analyze the anomaly. *POUTA-base* is POUTA without HSG and MSS modules, that is, there is no refined step in the discriminative network and the coarse information from reconstructive network is directly used to locate the anomaly. *POUTA-base+HSG* and *POUTA-base+MSS* add only HSG or MSS module to *POUTA-base* respectively. And *POUTA* adds both HSG and MSS modules to *POUTA-base*.

- 1) **Reconstruction errors vs representation reusing:** Experiments *Vanilla* and *POUTA-base* in Table VII are conducted to verify the effectiveness of the representation reusing. The results shows that by reusing the reconstructive representations instead of the pixel-level reconstruction errors, the performance improves. This demonstrates that reusing representations from the reconstructive network provides more accurate anomaly information than the conventional reconstruction errors.
- 2) **HSG module:** To investigate the effectiveness of HSG, experiments *POUTA-base* and *POUTA-base+HSG* are

conducted. *POUTA-base+HSG* adds only HSG module to *POUTA-base*, and the results show improvement in terms of the detection performance in Table VII. This shows that HSG module can be used to refine anomaly information. However, we also find that adding HSG only has marginal performance improvement. Due to the lack of MSS, the representations of the encoder and the decoder cannot represent the features of the original and reconstructed image fully. The semantic differences between the last few encoder representations and the first few decoder representations might not be obvious enough, resulting in unclear information about the abnormal regions at the high-level semantics, which affects the ability to guide the low-level representations.

- 3) **MSS module:** *POUTA-base+MSS* adds MSS module to *POUTA-base*. As shown in Table VII, the detection performance increases by adding MSS module. MSS requires each layer of discriminative network to predict the anomaly precisely, which enhances the ability to distinguish the normal and abnormal regions. While without HSG, each layer predicts the anomaly based on the representations of the current scale, which might be not robust and accurate enough since it has not been calibrated by the high-level semantic information. Therefore, adding only MSS module provides limited improvement.
- 4) **The coarse-to-fine process:** *POUTA* adds both HSG and MSS modules to *POUTA-base*. There is a significant performance increase as shown in Table VII. *POUTA* also shows a noticeable improvement when compared with *POUTA-base+HSG* and *POUTA-base+MSS*. The above results demonstrate that the collaboration between HSG and MSS plays an important role in refining the semantic information about the anomaly, allowing a better performance than adding only one or neither module.
- 5) **Parameters, computation and inference time.** We calculate the average inference time (including the time for deep network, and time for feature matching if it is an embedding-based method) per image on the embedding-based method *PatchCore* [16], the improved embedding-based method RD [18], the reconstruction-based method *Vanilla*, and the proposed method *POUTA*. Also, the computational cost (FLOPs and parameters of each model) are calculated. Calculation and parameters are metrics for network, while half of the procedure in *PatchCore* [16] is not completed by network, thereby only the inference time of [16] are measured and compared. The results are shown in Table VIII. The parameters of *POUTA* are obviously less than those of *Vanilla*, while the computation is slightly less than that of *Vanilla*. This is because *POUTA* utilizes the existing information more extensively and refines it. It can be seen from the inference time that *POUTA* is significantly more efficient than *Vanilla*, and marginally more efficient than *PatchCore*. Despite the floating point operations (FLOPs) of *POUTA* are slightly higher than those of the improved embedding-based method

TABLE VIII

THE CALCULATION COST (FLOPS), PARAMETERS AND AVERAGE INFERENCE TIME PER IMAGE OF VANILLA RECONSTRUCTION-BASED METHOD (SAME ARCHITECTURE WITH DR_{EM} [5]), EMBEDDING-BASED(PATCHCORE [16]), IMPROVED EMBEDDING-BASED METHOD (RD [18]), AND POUTA. THE BEST RESULT IS HIGHLIGHTED IN BOLD

	POUTA	Vanilla	Embedding-based[16]	RD[18]
Calculation	35.37G	39.38G	-	29.81G
Parameters	15.02M	28.68M	-	150.64M
Inference time	32ms	52ms	38ms	30ms

RD [18], there is no discernible difference in inference times between *POUTA* and RD, and the parameters of *POUTA* are considerably less than those of RD. Besides, *POUTA* demonstrates superior performance in terms of accuracy, as shown in Table I and II.

2) **Qualitative Results:** The qualitative results are exhibited in Figure 13. Also, we take Grad-CAM [4] to visualize the semantic differences captured by the five groups of models in Figure 13.

- 1) **Reconstruction errors vs representation reusing:** Figure 13 demonstrates that the reconstructive representations enable the model to capture more accurate anomaly semantics, resulting in *POUTA-base* to locate the anomaly more completely and precisely than *Vanilla*.
- 2) **HSG module:** As shown in Figure 13, the visualization of Grad-CAM shows that the addition of HSG enables the model to capture more accurate semantic differences, leading to predicting anomalies more precisely. This indicates that by introducing the high-level semantic information to the low-level feature through HSG module, it does benefit to calibrate the semantics. But the gains are limited, since without MSS, the high-level semantic difference is not clear enough to describe the anomaly, which weakens the ability to guide the low-level representations.
- 3) **MSS module:** As exemplified in Figure 13, it *POUTA-base+MSS* alleviates the misjudgment problem in *POUTA-base* through MSS module. We take t-SNE [42] to visualize the representation distribution before and after mapping layer (the last encoder representations and the first decoder representations) in the reconstructive network with or without the MSS module, as shown in Figure 14. It can be seen that, the main differences exist in the distributions of the encoder, where the encoder representations are significantly more separable with MSS, leading to a more obvious distribution shift between the encoder and decoder. The above phenomena verifies that MSS effectively improves the ability to distinguish the normal and abnormal information. However without HSG, the anomaly predictions of *POUTA-base+MSS* are still not ideal enough in some cases as shown in Figure 13, indicating that even if MSS module is added, HSG module is still necessary to calibrate the semantics.
- 4) **The coarse-to-fine process:** As presented in Figure 13, the anomaly predictions are significantly more precisely after adding HSG and MSS modules simultaneously.

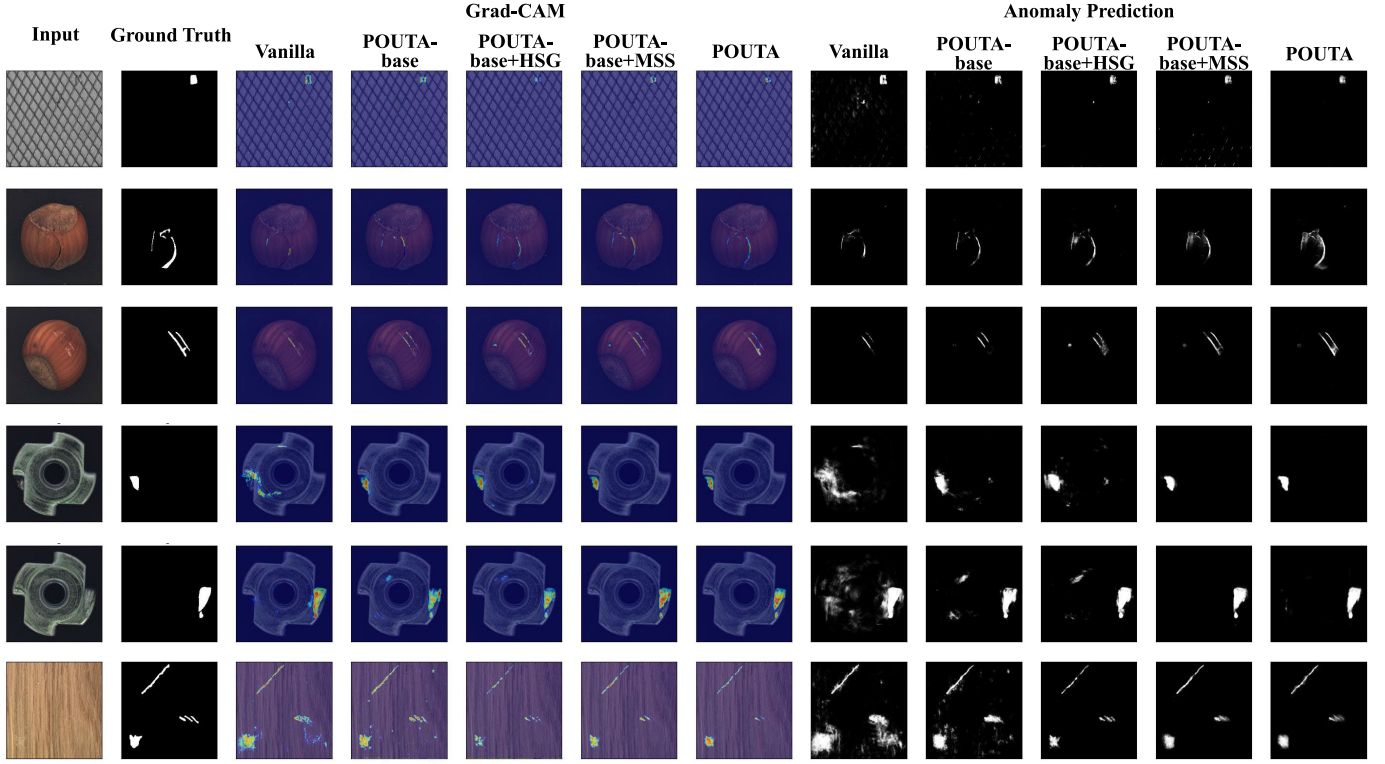


Fig. 13. The qualitative examples and visualizations (provided by Grad-CAM [4]) of vanilla method, POUTA-base and POUTA. POUTA obtains a more comprehensive set of anomaly information to locate the anomaly more precisely.

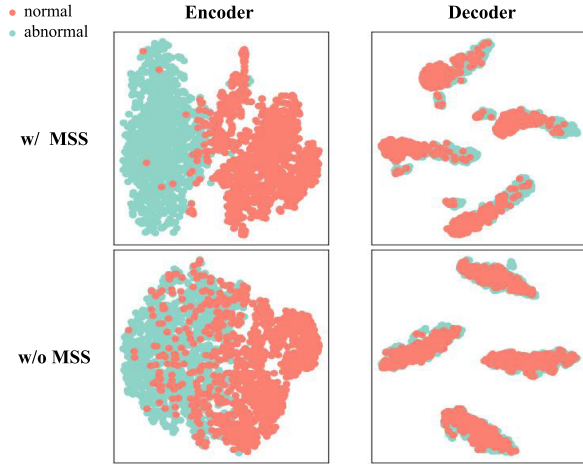


Fig. 14. We take t-SNE [42] to visualize the representation distribution before and after mapping layer (the last encoder representations and the first decoder representations) in reconstructive network under with or without MSS module. The red points correspond to the representations of the abnormal regions, and the green ones refer to the normal representations.

With HSG, the semantic information of each layer in the discriminative network is more accurate, since it has been calibrated by the high-level features, further enabling MSS to achieve better supervision effect. And MSS enables the distribution discrepancies between encoder and decoder to be more obvious and makes the high-level semantic anomaly information to be more clear by the supervision loss. The collaboration between HSG and MSS refines the semantic information about

the anomaly, allowing a significantly better performance than adding only one or neither module.

V. LIMITATION AND FUTURE POTENTIALS

As depicted in Figure 15, POUTA fails in some cases. The two cables placed on the bottom are the blue one on the left and the gray one on the right in the training data, but this input reverses their positions. POUTA points out the reversion and marks it on the predicted anomaly map. While the ground truth does not consider this to be an anomaly. POUTA shows the capability to identify numerous subtle anomalies, as indicated in Figure 8. While it is possible for POUTA to provide incomplete predictions when the anomaly is very insignificant, as presented in the second and third columns in Figure 15. It is possible to happen, since the minor anomalies always locate closely to the decision boundary, and the inconsistency between the fitted decision boundary and the real decision boundary is almost unavoidable. In the toothbrush shown in Figure 15, including the abnormal regions identified by the ground truth, POUTA also highlights additional regions where the bristles appear disorganized. But ground truth considers them as normal. And it is the same with the chewing gum in Figure 15, POUTA also highlights the bumpy regions, which are not considered as anomalies by the ground truth.

We propose POUTA as a lightweight and efficient model for the regular anomaly detection. POUTA demonstrates a strong capability to learn the inherent distribution of normal data, even in the few-shot setting. On the other hand, we find that POUTA is not suitable for the general anomaly detection (GAD). General anomaly detection aims to train a single

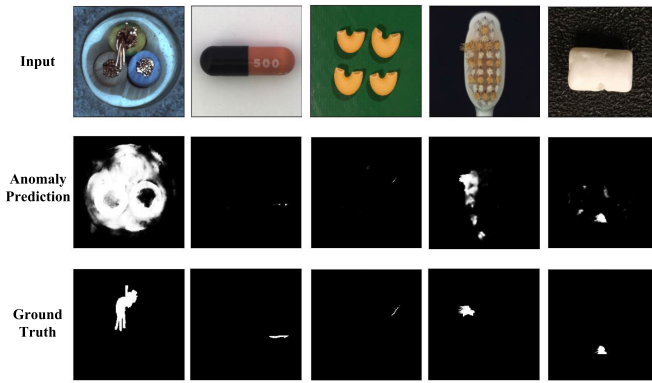


Fig. 15. Examples of failure cases.

anomaly detection model that can generalize to identifying anomaly in various datasets [65], [66], [67]. In general, there are a large number of parameters in the GAD model, which are much larger than those in POUTA. As a lightweight model, it is challenging for POUTA to accurately build the normal appearances of numerous datasets in a single model.

The anomaly detection towards the object in 3-dimension (3D) recently draws much attention. By replacing the reconstructive network with a 3D-aware network, it is possible to achieve 3D anomaly detection through POUTA.

VI. CONCLUSION

In this paper, we verifies that reusing the reconstructive representations is more reliable than the pixel-level reconstruction errors for reconstruction-based anomaly detection. Then a new approach named POUTA is proposed locating the anomaly by analyzing the distribution shift between the reconstructive encoder and decoder, which enables to predict the anomaly more precisely with lower cost than the vanilla method. Moreover, POUTA further refines the anomaly information by HSG and MSS modules in each discriminative layer, which also enhances the separability between the symmetric reconstructive representations. Finally, POUTA outperforms the prior methods on MVTEC AD, VisA and DAGM dataset, and achieves a comparable results with the best-performing supervised methods on DAGM dataset. Surprisingly, POUTA also achieves better performance than the state-of-the-art few-shot anomaly detection methods without any special design, showing that POUTA has strong ability to learn representations inherent in the training data.

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